

CLE-HFL evidence chain and information boundary

TRAIN-TIME PATH

CONTROLLED INPUT

1 CLE-HFL private data

client-specific class-operator binding

OBSERVED FAILURE

2 Shortcut-prone local training

robust local learning can still exploit binding

LOCAL MITIGATION

5 PEW + BER

coarse family witness + within-class support balance

SEALED EVALUATION AND ATTRIBUTION

REPORTING ONLY

3 Paired operator grid

same source and label; intervene on operator

TARGET-ALIGNED

4 DSA + local-first attribution

binding-directed mass shift; HFL versus Local

BOUNDARY-AWARE

6 Matched validation

ERM, CVaR, shared-PEW GroupDRO, maps/bases

motivates local target